

# SYAM SUNDAR REDDY N

AI Engineer | GenAI & LLM Applications (RAG, Agents) | Python Backend Engineer  
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## SUMMARY

Software engineer with ~4 years of industry experience transitioning into AI Engineering through hands-on development of end-to-end Generative AI systems. Builds RAG applications, agentic workflows, and text-to-SQL solutions using Python, FastAPI, LangChain, LangGraph, PostgreSQL/pgvector, Redis, Docker, and AWS, with emphasis on retrieval quality, evaluation, security, reliability, and production deployment.

## SKILLS

**LLM & Agents:** LLMs, Prompt Engineering, RAG, LangChain, LangGraph, AI Agents, MCP, Function Calling, Structured Outputs  
**Retrieval & Evaluation:** Embeddings, PostgreSQL/pgvector, Semantic Search, Hybrid Search (BM25), Reranking, Metadata Filtering, RAG Evaluation  
**Backend:** Python, FastAPI, REST APIs, SQL, PostgreSQL, SQLAlchemy, Redis, JWT, RBAC, Pytest  
**Infrastructure:** Docker, Git/GitHub, GitHub Actions, AWS (EC2, S3, RDS, ECR, CloudWatch), Logging, Monitoring, System Design

## WORK EXPERIENCE

- Taskora Private Limited** | Data Scientist Mar 2025 - Present
- Contributed to enterprise software delivery by translating business requirements into reliable application workflows and supporting implementation, testing, troubleshooting, and release activities.
  - Worked with engineering and product teams on application enhancements, production issue resolution, and operational improvements across business-critical workflows.
- Innovious IT Services Pvt. Ltd.** | Software Engineer Sep 2022 - Jan 2025
- Contributed to enterprise software development and maintenance across application enhancements, database-driven workflows, defect resolution, testing, documentation, and release activities.
  - Collaborated with cross-functional teams to analyze requirements, troubleshoot production issues, review code changes, and deliver software updates within agile development cycles.

## AI ENGINEERING PROJECTS

- Enterprise Knowledge & RAG Platform** FastAPI | LangChain | PostgreSQL/pgvector | Redis | Docker | AWS
- Built an end-to-end document ingestion pipeline covering parsing, chunking, metadata enrichment, embedding generation, and pgvector indexing for enterprise knowledge retrieval.
  - Implemented semantic retrieval, BM25 hybrid search, metadata filtering, reranking, source citations, and an evaluation harness for retrieval and answer-quality testing.
  - Added authenticated FastAPI endpoints, Redis caching, automated tests, Docker packaging, GitHub Actions CI/CD, monitoring, and AWS deployment.
- Agentic Customer Support Platform** LangGraph | MCP | FastAPI | PostgreSQL | Redis | LLMs
- Built a stateful support agent that orchestrates knowledge retrieval, customer lookup, ticket actions, and escalation through LangGraph workflows and MCP-based tools.
  - Used structured outputs and deterministic routing for tool selection, with persistent state, retries, validation, audit logging, and human-in-the-loop approval for sensitive actions.
  - Added automated scenario evaluation, authenticated APIs, containerized services, logging/monitoring, and a public cloud deployment.
- Enterprise SQL AI Agent** Python | LangGraph | FastAPI | PostgreSQL | Docker
- Built a schema-aware natural-language-to-SQL workflow with relevant-schema selection, structured SQL generation, validation, execution, retry handling, and result formatting.
  - Implemented read-only database access, destructive-query safeguards, authenticated APIs, query timeouts, benchmark tests, and automated failure-case evaluation.
  - Packaged the service with Docker and automated tests, then deployed the application for end-to-end demonstration and interview walkthroughs.

## EDUCATION

- Srinivasa Institute of Technology & Sciences** | Bachelor of Technology 2019 - 2022  
**Government Polytechnic, Rayachoty** | Diploma 2016 - 2019